

Exp. No: 3 Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

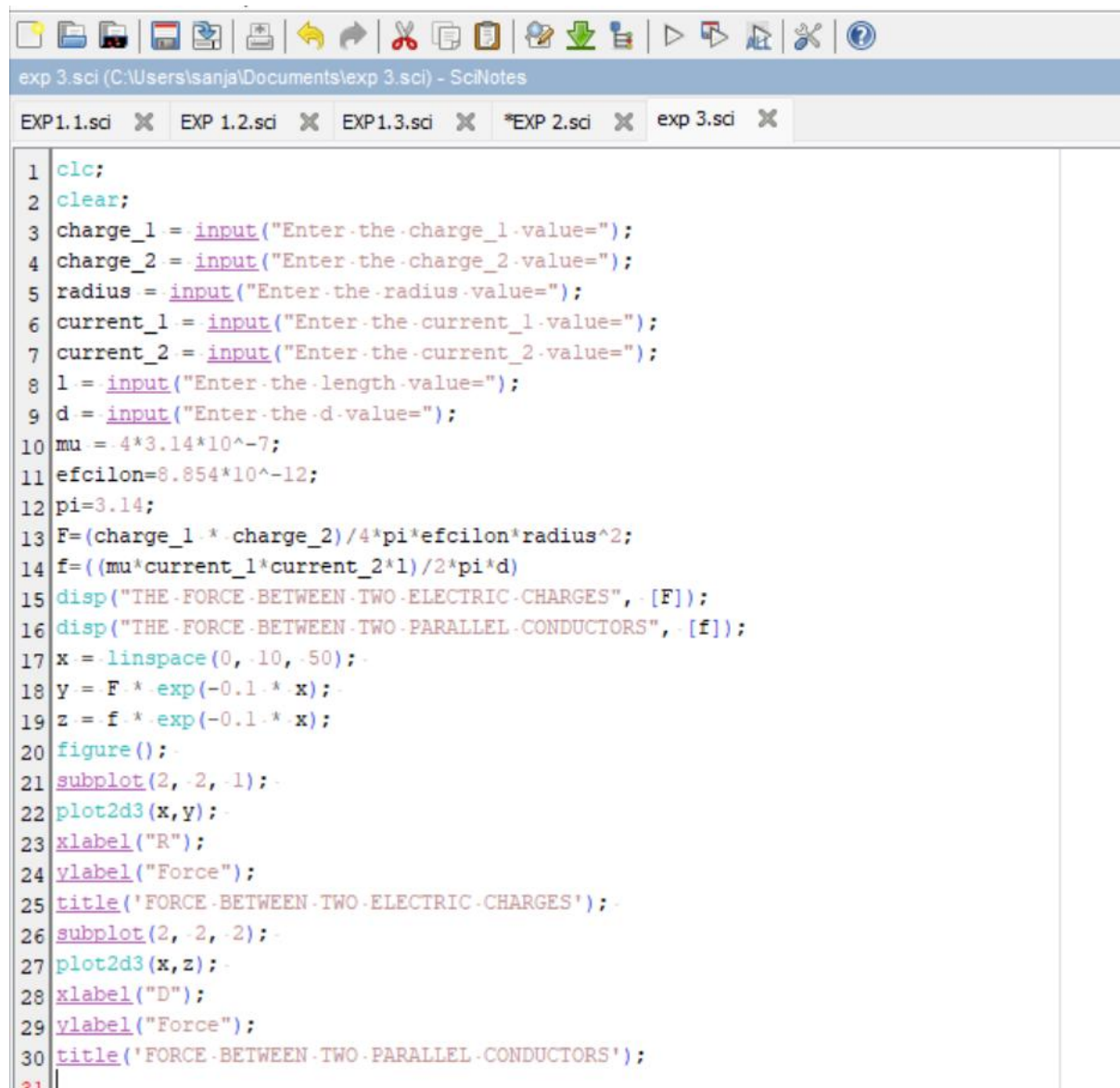
Aim:

To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

Software required:

- SCILAB version 6.0.1

Practical calculation:



```
exp 3.sci (C:\Users\sanja\Documents\exp 3.sci) - SciNotes
EXP1.1.sci X EXP 1.2.sci X EXP1.3.sci X *EXP 2.sci X exp 3.sci X
1 clc;
2 clear;
3 charge_1 = input("Enter the charge_1 value=");
4 charge_2 = input("Enter the charge_2 value=");
5 radius = input("Enter the radius value=");
6 current_1 = input("Enter the current_1 value=");
7 current_2 = input("Enter the current_2 value=");
8 l = input("Enter the length value=");
9 d = input("Enter the d value=");
10 mu = 4*3.14*10^-7;
11 efcilon=8.854*10^-12;
12 pi=3.14;
13 F=(charge_1 * charge_2)/4*pi*efcilon*radius^2;
14 f=((mu*current_1*current_2*l)/2*pi*d)
15 disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES", [F]);
16 disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS", [f]);
17 x = linspace(0, 10, 50);
18 y = F * exp(-0.1 * x);
19 z = f * exp(-0.1 * x);
20 figure();
21 subplot(2, 2, 1);
22 plot2d3(x, y);
23 xlabel("R");
24 ylabel("Force");
25 title('FORCE BETWEEN TWO ELECTRIC CHARGES');
26 subplot(2, 2, 2);
27 plot2d3(x, z);
28 xlabel("D");
29 ylabel("Force");
30 title('FORCE BETWEEN TWO PARALLEL CONDUCTORS');
```

File Edit Control Applications ?

File Browser C:\Users\sanja\Documents\ Scilab 2025.1.0 Console

Name

- Documents
- ..
- MATLAB
- My Music
- My Pictures
- My Videos
- cntrlsys2.sci
- cntrlsys3.sci
- cntrlsys4.sci
- cntrlsys5.sci
- cntrlsys6.1.sci
- cntrlsys6.sci
- cntrlsystems.1.sod
- CO318.sci
- CO3AT3.sci
- contlsys1.sci
- cs1.sci
- D.sci

Enter the charge_1 value= 1×10^{-6}

Enter the charge_2 value= 2×10^{-6}

Enter the radius value=3

Enter the current_1 value=2

Enter the current_2 value=4

Enter the length value=3

Enter the d value=4

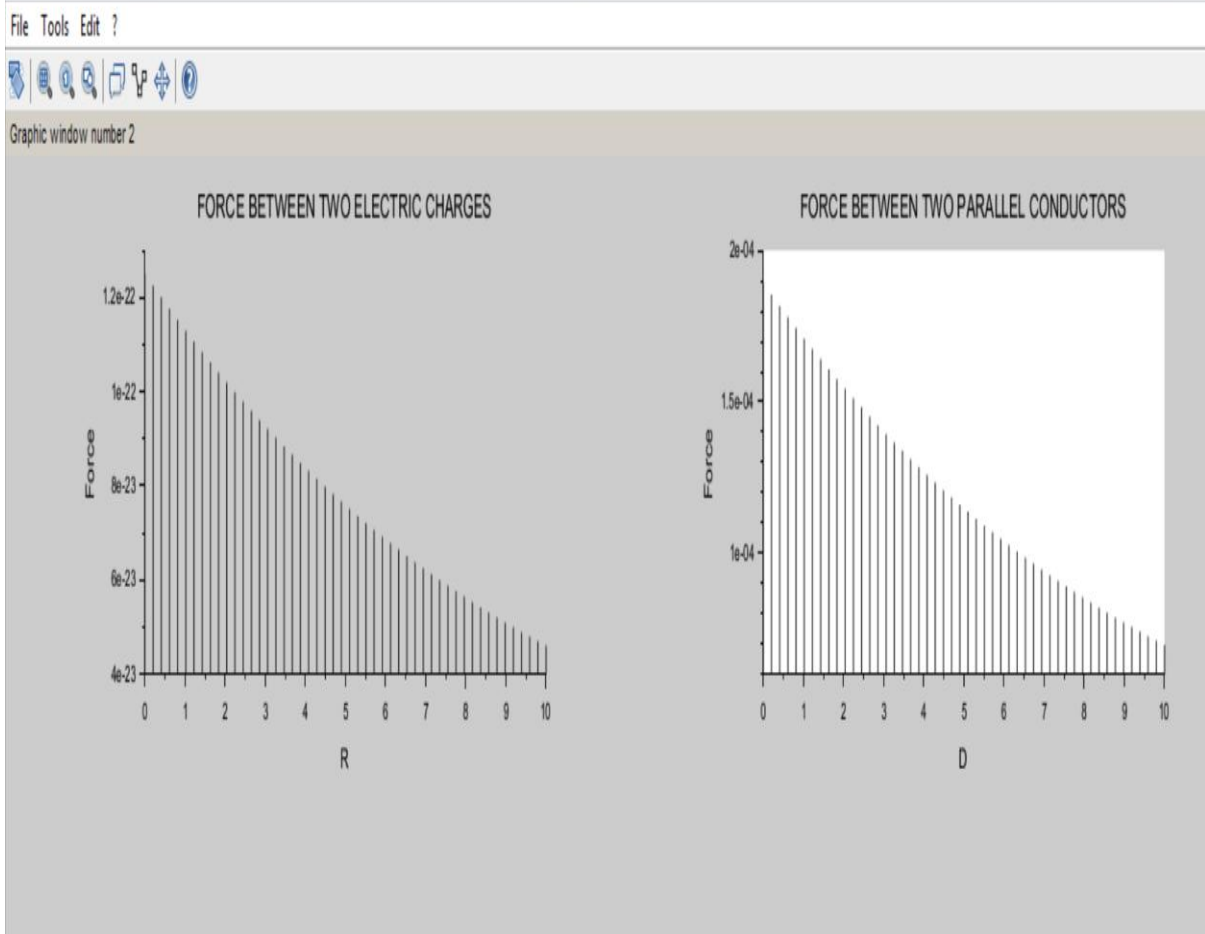
"THE FORCE BETWEEN TWO ELECTRIC CHARGES"

1.251D-22

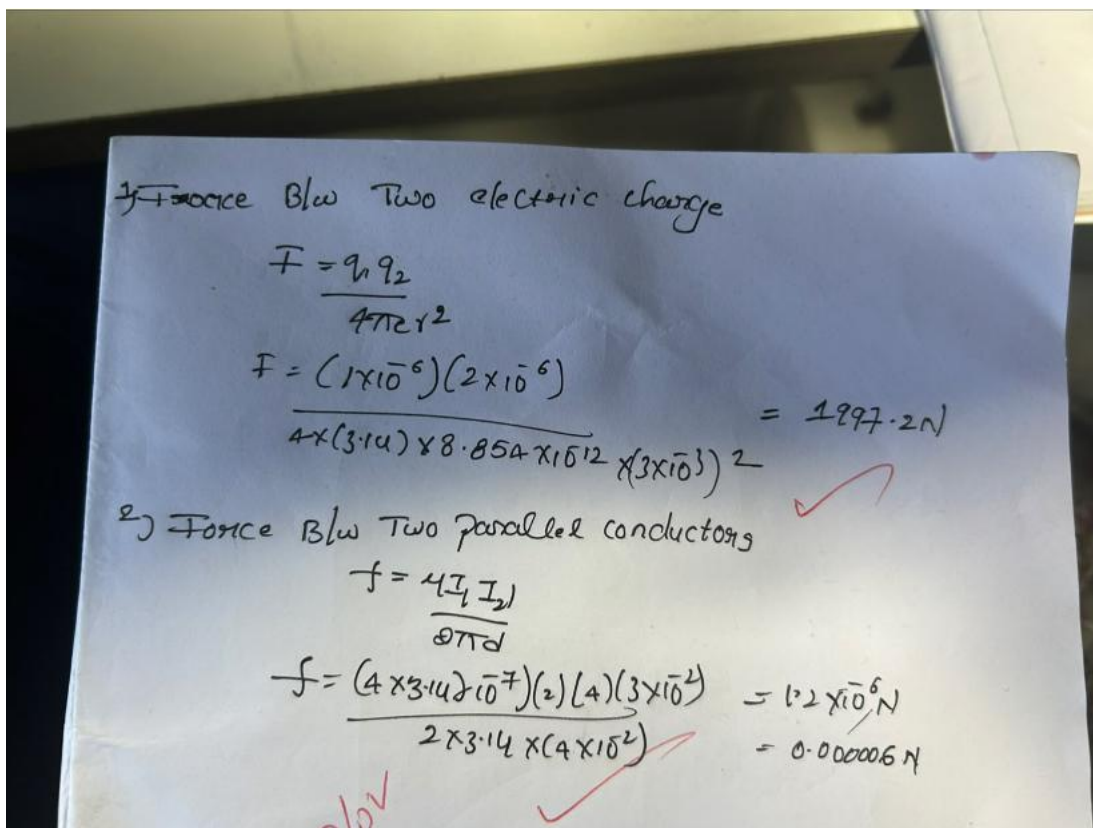
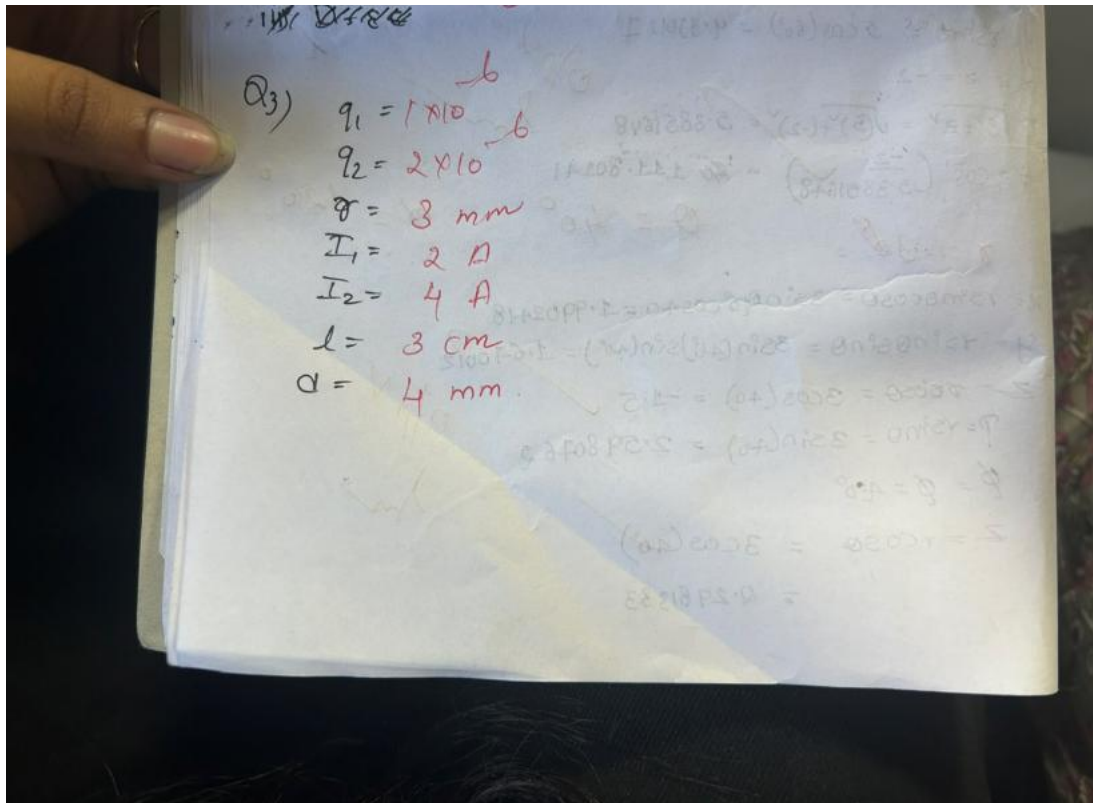
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"

0.0001893

Graphic window number 2 - 0



Theoretical calculation:



RESULTS:

The force of attraction between two electric charges and two parallel conductors was successfully determined and plotted for different separation distances. It was observed that the force decreases as the distance between the charges or conductors increases, following an inverse-square relationship. The plotted results verified the theoretical relationship between force and separation distance.